

Style guide / Common L^AT_EX mistakes

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1 Cross-references and citations

Label naming conventions

Labels should:

- be lowercase
- use hyphens, not spaces or camelCase
- include a type prefix (`sec:`, `fig:`, `tab:`, `eq:`, `thm:`, etc.)

✓ Good:

- `\label{sec:methods-derivation}`
- `\label{fig:simulation-results}`
- `\label{eq:score-function}`

✗ Bad:

- `\label{Section:Methods}`
- `\label{simulationResults}`
- `\label{Table 1}`

Use `\eqref` for equations

When referring to an equation, use `\eqref{eq:score-function}`, not `\ref{eq:score-function}`.

✓ Good:

`equation~\eqref{eq:score-function}`

equation (1.1)

✗ Bad:

`equation~\ref{eq:score-function}`

equation 1.1

The `eqref` version formats the reference as it appears in the actual equation (here, that's with parentheses, but the point is that if you change how equations look, the reference will always match the equation).

$$u(\beta) = \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X} \beta \quad (1.1)$$

Use nonbreaking spaces in references

Use `Section~\ref{sec:methods}`, not `Section \ref{sec:methods}`. The `~` prevents awkward line breaks.

Where to put citations?

This can be complicated. In general, citations should go at the end of a sentence:

Penalized regression methods are widely used in high-dimensional settings (Hastie et al., 2009; Zou and Hastie, 2005).

Every once in a while, the structure of a sentence requires a mid-sentence citation:

The lasso induces sparsity (Tibshirani, 1996), whereas ridge regression shrinks coefficients continuously (Hoerl and Kennard, 1970).

In that sentence, if you were to put both citations at the end, it would be ambiguous and confusing. But try to do this only when absolutely necessary, because it makes sentences less readable:

The lasso (Tibshirani, 1996) is widely used in high-dimensional regression (Hastie et al., 2009) and has inspired numerous extensions (Zou and Hastie, 2005; Fan and Li, 2001).

This sentence is choppy and hard to parse.

`\citep` versus `\citet`

Use `\citep{}` for parenthetical citations and `\citet{}` when the author name appears grammatically in the sentence.

✓ Good:

Several approaches have been proposed
`\citep{Fan2001, Zou2005}`. `\`

`\citet{Zou2005}` introduced the adaptive lasso.

Several approaches have been proposed (Fan and Li, 2001; Zou and Hastie, 2005).

Zou and Hastie (2005) introduced the adaptive lasso.

✗ Bad:

In `\citep{Zou2005}`, the adaptive lasso was proposed. `\`

Zou `\citep{Zou2005}` proposed the adaptive lasso.

In (Zou and Hastie, 2005), the adaptive lasso was proposed.

Zou (Zou and Hastie, 2005) proposed the adaptive lasso.

Incorrect capitalization in BibTeX

BibTeX styles often convert article titles to sentence case. Braces protect capital letters and acronyms that must remain capitalized.

✓ Good:

`title = {Analysis of {RNA}-Seq Data}`

Analysis of RNA-seq data

✗ Bad:

`title = {Analysis of RNA-Seq Data}`

Analysis of rna-seq data

2 Mathematical writing and typesetting

General advice

- Avoid double-subscripts like A_{n_i}
- Avoid introducing symbols for quantities you only use once
- When defining something, it is usually a good idea to give it both a technical and conceptual definition. For example: Let $L(\beta) = \sum_i \ell_i(\beta)$ denote the log-likelihood, which measures how well the value β agrees with the data.

Math mode is for mathematical symbols, not ordinary text

Unadorned letters in math mode are interpreted as variables. If you write `AIC` in math mode, this will be typeset as AIC : the variable A times the variable I times the variable C . If you need `AIC` in an equation, use `\text{AIC}`, `\text{CV}`, and so on.

Similarly, don't write `log`, `exp`, `max`, and so on. There are precisely defined commands for all these mathematical operators that preserve correct spacing and typography: `\log`, `\exp`, `\max`.

✓ Good:

```
$ \text{AIC} = -2\ell + 2p $ \\
$ \exp x = \max_y \log y $
```

$$\text{AIC} = -2\ell + 2p$$

$$\exp x = \max_y \log y$$

✗ Bad:

```
$ AIC = -2\ell + 2p $ \\
$ \exp x = \max_y \log y $
```

$$AIC = -2\ell + 2p$$

$$\exp x = \max_y \log y$$

Don't type the pipe

Use `\mid`, `\abs`, and `\norm`. Never type `|` directly.

✓ Good:

```
$ \abs{x} > \Pr(\norm{y} \mid z $)
```

$$|x| > \mathbb{P}(\|y\| \mid z)$$

✗ Bad:

```
$ |x| > \Pr(||y|| | z $)
```

$$|x| > \mathbb{P}(\|y\| | z)$$

Equations are part of the sentence

Equations should follow ordinary punctuation rules.

If the equation ends a sentence, it needs a period, like this:

```
$$ L(\beta) = \sum_{i=1}^n \ell_i(\beta). $$
```

If the equation ends a clause, as in

```
$$ L(\beta) = \sum_{i=1}^n \ell_i(\beta), $$
```

it needs a comma.

If the equation ends a sentence, it needs a period, like this:

$$L(\beta) = \sum_{i=1}^n \ell_i(\beta).$$

If the equation ends a clause, as in

$$L(\beta) = \sum_{i=1}^n \ell_i(\beta),$$

it needs a comma.

Displayed equations are not punctuation-free objects floating outside the grammar of the paragraph.

3 Typography and punctuation

Dashes and hyphens

- Use a single `-` for hyphenated words
- Use two dashes `--` for ranges and relationships
- Use three dashes `---` for interruptions and emphasis

```
Brad Efron is a well-known statistician. \\
The bias--variance tradeoff is shown on
pages 10--15. \\
Bias --- while traditionally considered bad ---
can often be useful.
```

Brad Efron is a well-known statistician.
The bias–variance tradeoff is shown on pages 10–15.
Bias — while traditionally considered bad — can often be useful.

LaTeX requires directional quotes

✓ Good:

```
`quoted text''
```

$$\text{“quoted text”}$$

✗ Bad:

```
"quoted text"
```

$$\text{”quoted text”}$$

Note that the quotes on the left are backwards in the “bad” example.

4 Miscellaneous

Do not ignore LaTeX errors and warnings

A document should compile cleanly before you share it with any collaborators. Almost all warnings indicate genuine problems: undefined references, doubly-defined references, missing citations, overfull boxes, package conflicts, etc. It doesn't matter if the PDF looks OK to you — the warning is there for a reason and usually indicates a serious underlying problem that will only snowball, growing worse and worse the longer you take to fix it.

Don't say “real data”

Never title a section “Real Data”. This tells the reader nothing about the application. Every dataset used in a scientific study is real. The purpose of an application is to demonstrate how the method addresses a substantive scientific question. Use descriptive titles and engage with the scientific context of the analysis.

✓ Good:

In Section 5, we apply our method to a gene expression study of breast cancer patients.

✗ Bad:

In Section 5, we apply our method to real data.

5 Figures and tables

Every figure and table should:

- have a caption
- have a label
- be referenced in the text (using the label, not something like “see the figure below”)

Figures

The call to `\includegraphics{}` should not include the directory path — only the file name. The path to the directory where the figures are located should be set with `\graphicspath{}` in the preamble; see the template for an example.

```
\begin{figure}  
  \includegraphics[width=\linewidth]{t-fig.pdf}  
  \caption{Student's  $t$ -test (which assumes  
    equal variance) suffers from a lack of power  
    when the large-variance group is has more  
    observations than the small-variance group.}  
  \label{fig:t-variance}  
\end{figure}
```

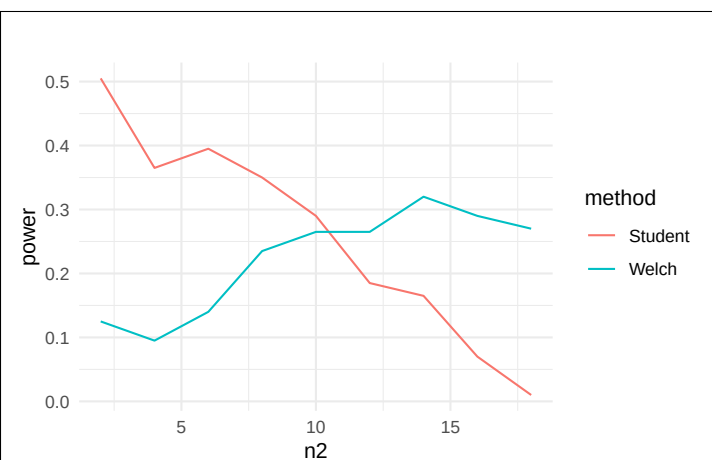


Figure 0: Student's t -test (which assumes equal variance) suffers from a lack of power when the large-variance group is has more observations than the small-variance group.

Tables

1. Use `\toprule`, `\midrule`, and `\bottomrule` (from the `booktabs` package)
2. Never use vertical rules.

3. Align numbers consistently.
4. Report only enough digits to convey meaningful information. No one could possibly care that the type 1 error rate was 0.1121 instead of 0.11. Every digit must justify its existence. Extra digits consume space, distract the reader, and almost never affect the scientific conclusion. When in doubt, round more aggressively.

For more on how to lay out a professional-looking table, see the [booktabs documentation](#).

```
\begin{table}
\centering
\caption{Type 1 error for Student and Welch
 $t$ -tests. Ratio ( $n_1:n_2$ ) displays the sample
size of the low-variance and high-variance
groups (low-variance first). Student's test
suffers from inflated Type I error when the
small-variance group is larger.}
\label{tab:simulation-results}

\begin{tabular}{lrr}
\toprule
Ratio & Student & Welch\\
\midrule
4:16 & 0.03 & 0.06\\
10:10 & 0.05 & 0.05\\
16:4 & 0.14 & 0.06\\
\bottomrule
\end{tabular}
\end{table}
```

Table 0: Type 1 error for Student and Welch t -tests. Ratio ($n_1 : n_2$) displays the sample size of the low-variance and high-variance groups (low-variance first). Student's test suffers from inflated Type I error when the small-variance group is larger.

Ratio	Student	Welch
4:16	0.03	0.06
10:10	0.05	0.05
16:4	0.14	0.06

Tables should be reproducible

The above example is somewhat misleading: in most cases, you shouldn't be directly typing the table contents. The table contents should be automatically generated, typically from `knitr::kable("latex", booktabs = TRUE)`, then use `\input{}`, as in the following, where the output from `kable()` has been saved as `t-tab.tex`:

```
\begin{table}
\centering
\caption{Type 1 error for Student and Welch
 $t$ -tests. Ratio ( $n_1:n_2$ ) displays the sample
size of the low-variance and high-variance
groups (low-variance first). Student's test
suffers from inflated Type I error when the
small-variance group is larger.}
\label{tab:same-simulation-results}
\input{t-tab}
\end{table}
```

Table 0: Type 1 error for Student and Welch t -tests. Ratio ($n_1 : n_2$) displays the sample size of the low-variance and high-variance groups (low-variance first). Student's test suffers from inflated Type I error when the small-variance group is larger.

Ratio	Student	Welch
4:16	0.03	0.06
10:10	0.05	0.05
16:4	0.14	0.06

The output is exactly the same, but the point is that if you alter or re-run the simulation, the document is guaranteed to track those changes.

References

- FAN, J. and LI, R. (2001). Variable selection via nonconcave penalized likelihood and its oracle properties. *Journal of the American Statistical Association*, **96** 1348–1360. URL doi.org/10.1198/016214501753382273.
- HASTIE, T., TIBSHIRANI, R. and FRIEDMAN, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer.
- HOERL, A. E. and KENNARD, R. W. (1970). Ridge regression: applications to nonorthogonal problems. *Technometrics*, **12** 69–82.
- TIBSHIRANI, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B*, **58** 267–288.
- ZOU, H. and HASTIE, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society Series B*, **67** 301–320.